Infosys - Coding 18

HackerRank

You have N items, and the weight of the i-th item is A[i] kg.

You need to distribute these N items into bags such that:

• Each bag can hold at most K kg in total.

• Each item must be placed in exactly one bag.

• Items cannot be divided into smaller pieces.

You will find the minimum number of bags (Q) needed to distribute all the items under these constraints. Find the value of (Q × K).

Input Format

The first line contains an integer, N, denoting the number of elements in A.

The next line contains an integer, K, denoting the maximum weight for a bag.

Each line i of the N subsequent lines (where 0 ≤ i < N) contains an integer describing A[i].

Constraints

1 <= N <= 10

1 <= K <= 1000

1 <= A[i] <= 1000

Output Format

The output contains an integer value

Sample Input 0

```
2
8
3
4
```

Sample Output 0

```
8
```

Explanation 0

N=2, K=8, A=[3 4] One bag can contain all items because k = 8 > 3+4, so the answer is 1 * k = 8.

Sample Input 1

```
2
8
4
5
```

Sample Output 1

```
16
```

Explanation 1

N=2, K=8, A=[4 5] We need two bags because 4+5 > 8, so the answer is 2 * k = 16.